

SANJAY M

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Summary

AI & Data Science undergraduate skilled in building AI-powered intelligence and decision-support systems using NLP, LLMs, and Retrieval-Augmented Generation.

Technical Skills

Artificial Intelligence: Machine Learning, NLP, Transformer, CNN, LSTM

Generative AI Stack: Retrieval-Augmented Generation, LLMs, LangChain, Vector Database, Hugging Face

Full-Stack Development: React.js, FastAPI, PostgreSQL, MySQL

Programming & Tools: Java, Python, SQL, GitHub, Postman

Projects

AskMe — AI-Powered Document Q&A / Intelligent Policy Advisor

TypeScript · Python · LangChain · FAISS · Vector Database · Llama 3 (LLM) · NLP · Retrieval-Augmented Generation (RAG) · PyMuPDF

github.com/SanjayMuthuswamy/AskMe

- Engineered a production Retrieval-Augmented Generation (RAG) pipeline enabling users to upload PDFs and receive instant, source-grounded answers via semantic search and large language model (LLM) inference.
- Designed a hybrid retrieval architecture combining FAISS vector search with BM25 keyword search and a cross-encoder reranker, achieving Precision@5: 0.833, nDCG@5: 0.908, MRR@10: 0.889, with 0% hallucination.

CompeteTracker — AI-Powered Competitive Intelligence Agent

TypeScript · Python · NLP · Llama 3 (LLM) · Agentic AI · MySQL · BeautifulSoup · Web Scraping · REST API

github.com/SanjayMuthuswamy/CompeteTracker

- Architected an AI-native, agentic intelligence platform that automatically monitors competitors' product launches, feature updates, and marketing activity across multiple online sources.
- Built an automated data pipeline combining web scraping (BeautifulSoup), NLP, and LLM-based summarization, synthesizing findings into structured, decision-ready intelligence reports backed by a normalized MySQL schema.

DeepFakeGuard — AI-Powered Deepfake Detection Platform

JavaScript · Python · TensorFlow · Keras · Deep Learning · Computer Vision · CNN · LSTM · EfficientNet-B0

github.com/SanjayMuthuswamy/DeepFakeGuard

- Designed and trained a CNN + LSTM hybrid deep learning model on EfficientNet-B0 to detect manipulated regions in images and video frames with high classification accuracy.
- Applied computer vision and deep learning techniques (TensorFlow/Keras) to build a full-stack web interface delivering real-time, frame-level manipulation analysis to end users.

Open Source Contribution

Open Source Contributor — Hugging Face (PEFT)

- Improved official documentation for VeRA in the PEFT library, clarifying parameter-efficient fine-tuning concepts to make them easier for developers to understand and adopt.
- Participated in the open-source review process end-to-end, addressing maintainer feedback and merging changes into the Hugging Face ecosystem.

Certifications & Achievements

- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate** — Oracle
- Machine Learning, Deep Learning & NLP** — Udemy
- Finalist, AI Agent Hackathon 2025** — built CompeteTracker, an AI-native competitor intelligence platform.

Education

Bannari Amman Institute of Technology

Sep 2023 – Present

B.Tech in Artificial Intelligence and Data Science | **CGPA: 8.3 / 10**

Kongu Vellalar Matric Hr. Sec. School, Coimbatore

2019 – 2023

HSC — Class XII | **Percentage: 89.8%**

Languages

English (Professional) | Tamil (Native)